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#include <iostream>

#include <queue>

using namespace std;

#define V 6 // Number of landmarks

// Landmark names
string landmarks[V] = {
    "College Gate", "Library", "Canteen",
    "Playground", "Hostel", "Parking"
};

// ----- DFS using Adjacency Matrix -----
void DFS(int adjMatrix[V][V], int start, bool visited[]) {
    cout << landmarks[start] << " -> ";
    visited[start] = true;

    for (int i = 0; i < V; i++) {
        if (adjMatrix[start][i] == 1 && !visited[i]) {
            DFS(adjMatrix, i, visited);
        }
    }
}

// ----- Node structure for Adjacency List -----
struct Node {
    int data;
    Node* next;
};

```

```

// Add edge to adjacency list
void addEdge(Node* adjList[], int u, int v) {
    // Add v to u's list
    Node* newNode = new Node;
    newNode->data = v;
    newNode->next = adjList[u];
    adjList[u] = newNode;

    // Since graph is undirected, add u to v's list
    newNode = new Node;
    newNode->data = u;
    newNode->next = adjList[v];
    adjList[v] = newNode;
}

```

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// ----- BFS using Adjacency List -----

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void BFS(Node* adjList[], int start) {
    bool visited[V] = {false};
    queue<int> q;

    visited[start] = true;
    q.push(start);

    while (!q.empty()) {
        int node = q.front();
        q.pop();

        cout << landmarks[node] << " -> ";
    }
}

```

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Node* temp = adjList[node];
while (temp != NULL) {
    int neighbor = temp->data;
    if (!visited[neighbor]) {
        visited[neighbor] = true;
        q.push(neighbor);
    }
    temp = temp->next;
}
}
}

```

// ----- Main Function -----

```
int main() {
```

```
// Adjacency Matrix
```

```
int adjMatrix[V][V] = {0};
```

```
// Adding edges
```

```
adjMatrix[0][1] = adjMatrix[1][0] = 1;
```

```
adjMatrix[0][2] = adjMatrix[2][0] = 1;
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```
adjMatrix[1][3] = adjMatrix[3][1] = 1;
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```
adjMatrix[2][4] = adjMatrix[4][2] = 1;
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```
adjMatrix[3][5] = adjMatrix[5][3] = 1;
```

```
adjMatrix[4][5] = adjMatrix[5][4] = 1;
```

```
// Adjacency List (array of pointers)
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```
Node* adjList[V];
```

```

// Initialize list
for (int i = 0; i < V; i++) {
    adjList[i] = NULL;
}

// Adding edges
addEdge(adjList, 0, 1);
addEdge(adjList, 0, 2);
addEdge(adjList, 1, 3);
addEdge(adjList, 2, 4);
addEdge(adjList, 3, 5);
addEdge(adjList, 4, 5);

// DFS
cout << "DFS Traversal (Adjacency Matrix):\n";
bool visited[V] = {false};
DFS(adjMatrix, 0, visited);

cout << "\n\n";

// BFS
cout << "BFS Traversal (Adjacency List):\n";
BFS(adjList, 0);

cout << endl;

return 0;
}

```